

Лабораторная работа

Номер 7

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Информация

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Получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

Создание нового проекта в GNS3

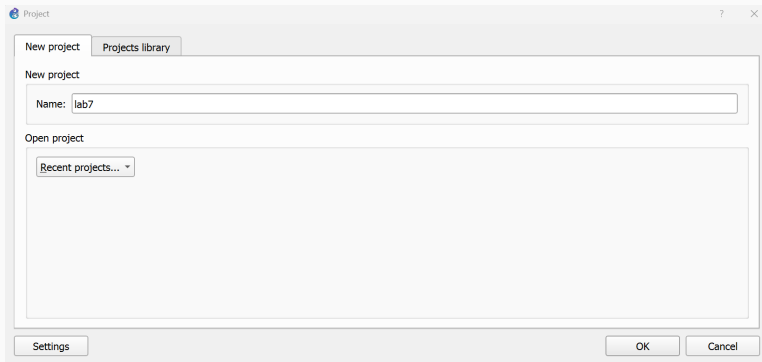


Рис. 1: Создание нового проекта в GNS3

Топология сети для настройки DHCPv4

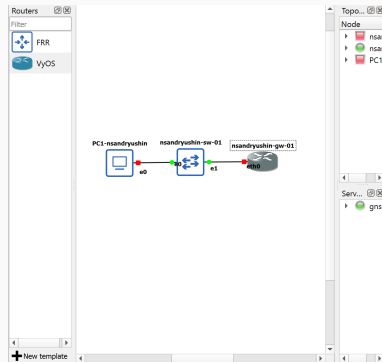


Рис. 2: Топология сети для настройки DHCPv4

Запуск захвата трафика в Wireshark

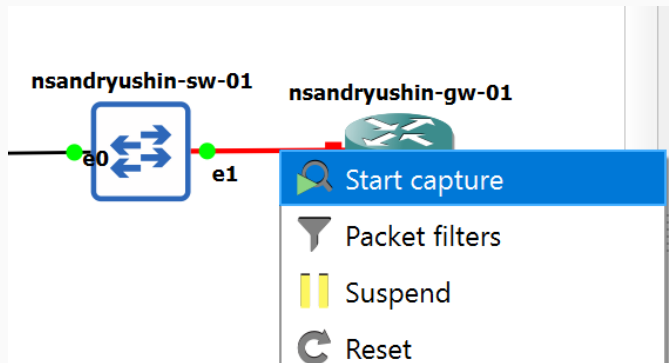


Рис. 3: Запуск захвата трафика в Wireshark

Базовая настройка маршрутизатора VyOS

```
vyos login: vyos
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name nsandryushin-gw-01
[edit]
vyos@vyos# set system domain-name nsandryushin.net
[edit]
vyos@vyos# set system login user nsandryushin authentication plaintext-password
123456
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
logout
```

Рис. 4: Базовая настройка маршрутизатора VyOS

Удаление стандартного пользователя

```
nsandryushin@nsandryushin-gw-01:~$ configure
[edit]
nsandryushin@nsandryushin-gw-01# delete system login user vyos
[edit]
nsandryushin@nsandryushin-gw-01# commit
[edit]
nsandryushin@nsandryushin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

Рис. 5: Удаление стандартного пользователя

Настройка IPv4 адресации и DHCP-сервера

```
nsandryushin@nsandryushin-gw-01:~$ configure
(edit)
nsandryushin@nsandryushin-gw-01# delete interfaces ethernet eth0 address dhcp
(edit)
nsandryushin@nsandryushin-gw-01# commit
(edit)
nsandryushin@nsandryushin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
(edit)
nsandryushin@nsandryushin-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
(edit)
nsandryushin@nsandryushin-gw-01# set service dhcp-server shared-network-name nsandryushin domain-name nsandryushin.net
(edit)
nsandryushin@nsandryushin-gw-01# set service dhcp-server shared-network-name nsandryushin name-server 10.0.0.1
(edit)
nsandryushin@nsandryushin-gw-01# set service dhcp-server shared-network-name nsandryushin subnet 10.0.0.0/24 default-router 10.0.0.1
(edit)
nsandryushin@nsandryushin-gw-01# set service dhcp-server shared-network-name nsandryushin subnet 10.0.0.0/24 range hosts start 10.0.0.2
(edit)
nsandryushin@nsandryushin-gw-01# set service dhcp-server shared-network-name nsandryushin subnet 10.0.0.0/24 range hosts stop 10.0.0.253
(edit)
nsandryushin@nsandryushin-gw-01# commit
(edit)
nsandryushin@nsandryushin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
(edit)
nsandryushin@nsandryushin-gw-01# exit
exit
nsandryushin@nsandryushin-gw-01:~$
```

Рис. 6: Настройка IPv4 адресации и DHCP-сервера

Проверка статистики DHCP-сервера

```
nsandryushin@nsandryushin-gw-01:~$ show dhcp server statistics
Pool          Size    Leases    Available  Usage
-----
nsandryushin   252      0         252       0%
nsandryushin@nsandryushin-gw-01:~$ show dhcp server leases
IP address    Hardware address  State    Lease start    Lease expiration  Re
maining      Pool      Hostname
-----
nsandryushin@nsandryushin-gw-01:~$
```

Рис. 7: Проверка статистики DHCP-сервера

Получение IP-адреса на клиенте PC1 (процесс DORA)

```

DHCP: to dhcp-2
Sender: 1 (REQUEST)
Client IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:10:1b:44:00:00
Option 53: Message Type = Discover
Option 12: Host Name = VCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:10:1b:44:00:00
Option 1 (REQUEST)
Client IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:10:1b:44:00:00
Option 53: Message Type = Offer
Option 12: Host Name = VCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:10:1b:44:00:00
Option 2 (OFFER)
Client IP Address: 0.0.0.0
Server IP Address: 10.0.0.2
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:10:1b:44:00:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 15: Lease Time = 36480
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 4: DNS Server = 10.0.0.1
Option 15: Domain = mskysanlin.net
Sender: 1 (REQUEST)
Client IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:10:1b:44:00:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
Option 50: Requested IP Address = 10.0.0.2
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:10:1b:44:00:00
Option 12: Host Name = VCS
Option 2 (REQUEST)
Client IP Address: 10.0.0.2
Server IP Address: 10.0.0.2
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:10:1b:44:00:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 11: Lease Time = 36480
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 4: DNS Server = 10.0.0.1
Option 15: Domain = mskysanlin.net
IP 10.0.0.2/24 on 10.0.0.1
VCS> done

```

Рис. 8: Получение IP-адреса на клиенте PC1 (процесс DORA)

Проверка списка выданных адресов (Leases)

```
nsandryushin@nsandryushin-gw-01:~$ show dhcp server statistics
Pool          Size      Leases    Available  Usage
-----
nsandryushin  252      1         251        0%
nsandryushin@nsandryushin-gw-01:~$ show dhcp server leases
IP address    Hardware address  State    Lease start      Lease expiration
  Remaining    Pool              Hostname
-----
10.0.0.2      00:50:79:66:68:00  active   2025/12/06 13:37:06  2025/12/07 13:37
:06 23:58:40      nsandryushin  VPCS
```

Рис. 9: Проверка списка выданных адресов (Leases)

Просмотр логов DHCP-сервера

```
nsandryushin@nsandryushin-gw-01:~$ show dhcp server statistics
Pool      Size      Leases    Available  Usage
-----
nsandryushin 252        1         251      0%
nsandryushin@nsandryushin-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start  Lease expiration
Remaining  Pool              Hostname
-----
10.0.0.2    00:50:79:66:68:00 active  2025/12/06 13:37:06 2025/12/07 13:37:06
23:58:40    nsandryushin VPCS
nsandryushin@nsandryushin-gw-01:~$ show log | grep dhcp
Dec 06 13:10:46 dhclient-script-vyos[1615]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Dec 06 13:10:47 vyos-hostsd[607]: Request data: {"type": "search_domains", "op": "delete", "data": ["dhcp-eth0"]}
Dec 06 13:10:47 dhclient-script-vyos[1615]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Dec 06 13:10:48 vyos-hostsd[607]: Request data: {"type": "name_servers", "op": "delete", "data": ["dhcp-eth0"]}
Dec 06 13:10:53 dhclient-script-vyos[1776]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Dec 06 13:10:55 vyos-hostsd[607]: Request data: {"type": "search_domains", "op": "delete",
```

Рис. 10: Просмотр логов DHCP-сервера

Детальный анализ логов работы DHCP

```
Dec 06 13:34:30 sudo[10576]: nsandryushin : TTY=ttys0 ; PWD=/home/nsandryushin ; USER=root  
; COMMAND=/usr/bin/sh -c /usr/sbin/vysshim /usr/libexec/vyos/conf_mode/dhcp_server.py  
Dec 06 13:34:30 vyos-configd[608]: Received message: {"type": "node", "data": "/usr/libexec  
/vyos/conf_mode/dhcp_server.py"}  
Dec 06 13:34:32 dhcpd[10593]: Wrote 0 leases to leases file.  
Dec 06 13:34:32 dhcpd[10593]: Lease file test successful, removing temp lease file: /config  
/dhcpd.leases.1765028072  
Dec 06 13:34:32 dhcpd[10595]: Wrote 0 leases to leases file.  
Dec 06 13:34:32 dhcpd[10595]:  
Dec 06 13:34:32 dhcpd[10595]: No subnet declaration for eth2 (no IPv4 addresses).  
Dec 06 13:34:32 dhcpd[10595]: ** Ignoring requests on eth2. If this is not what  
Dec 06 13:34:32 dhcpd[10595]: you want, please write a subnet declaration  
Dec 06 13:34:32 dhcpd[10595]: in your dhcpd.conf file for the network segment  
Dec 06 13:34:32 dhcpd[10595]: to which interface eth2 is attached. **  
Dec 06 13:34:32 dhcpd[10595]:  
Dec 06 13:34:32 dhcpd[10595]: No subnet declaration for eth1 (no IPv4 addresses).  
Dec 06 13:34:32 dhcpd[10595]: ** Ignoring requests on eth1. If this is not what  
Dec 06 13:34:32 dhcpd[10595]: you want, please write a subnet declaration  
Dec 06 13:34:32 dhcpd[10595]: in your dhcpd.conf file for the network segment  
Dec 06 13:34:32 dhcpd[10595]: to which interface eth1 is attached. **  
Dec 06 13:34:32 dhcpd[10595]:  
Dec 06 13:34:32 dhcpd[10595]: Server starting service.  
Dec 06 13:36:01 sudo[10672]: nsandryushin : TTY=ttys0 ; PWD=/home/nsandryushin ; USER=root  
; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --statistics  
Dec 06 13:36:10 sudo[10698]: nsandryushin : TTY=ttys0 ; PWD=/home/nsandryushin ; USER=root  
; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --leases  
Dec 06 13:37:02 dhcpd[10595]: DHCPDISCOVER from 00:50:79:66:68:00 via eth0  
Dec 06 13:37:03 dhcpd[10595]: DHCPPOFFER on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via eth0  
Dec 06 13:37:06 dhcpd[10595]: DHCPREQUEST for 10.0.0.2 (10.0.0.1) from 00:50:79:66:68:00 (V  
PCS) via eth0  
Dec 06 13:37:06 dhcpd[10595]: DHCPACK on 10.0.0.2 to 00:50:79:66:68:00 (VPCS) via eth0  
Dec 06 13:38:21 sudo[10724]: nsandryushin : TTY=ttys0 ; PWD=/home/nsandryushin ; USER=root  
; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --statistics  
Dec 06 13:38:25 sudo[10750]: nsandryushin : TTY=ttys0 ; PWD=/home/nsandryushin ; USER=root  
; COMMAND=/usr/libexec/vyos/op_mode/show_dhcp.py --leases  
nsandryushin@nsandryushin-qw-01:~$
```

Рис. 11: Детальный анализ логов работы DHCP

Анализ DHCP пакетов в Wireshark

2	0.030334	10.0.0.1	10.0.0.2	ICMP	62 Echo (ping) request id=0xef2a, seq=0/0, ttl=64 (reply in 3)
3	0.030813	10.0.0.2	10.0.0.1	ICMP	62 Echo (ping) reply id=0xef2a, seq=0/0, ttl=64 (request in 2)
4	1.000781	0.0.0.0	255.255.255.255	DHCP	406 DHCP Discover - Transaction ID 0x3540fe03
5	1.011014	0c:57:62:65:00:00	Broadcast	ARP	60 Who has 10.0.0.3? Tell 10.0.0.1
6	2.017867	10.0.0.1	10.0.0.3	DHCP	342 DHCP Offer - Transaction ID 0x3540fe03
7	2.021854	0c:57:62:65:00:00	Broadcast	ARP	60 Who has 10.0.0.3? Tell 10.0.0.1
8	3.047320	0c:57:62:65:00:00	Broadcast	ARP	60 Who has 10.0.0.3? Tell 10.0.0.1
9	4.002783	0.0.0.0	255.255.255.255	DHCP	406 DHCP Request - Transaction ID 0x3540fe03
10	4.054227	10.0.0.1	10.0.0.3	DHCP	342 DHCP ACK - Transaction ID 0x3540fe03
11	5.002995	Private_66:68:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.3 (Request)
12	5.222258	0c:57:62:65:00:00	Private_66:68:00	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
13	6.004398	Private_66:68:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.3 (Request)
14	6.245323	0c:57:62:65:00:00	Private_66:68:00	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
15	7.005024	Private_66:68:00	Broadcast	ARP	64 Gratuitous ARP for 10.0.0.3 (Request)
16	7.269790	0c:57:62:65:00:00	Private_66:68:00	ARP	60 Who has 10.0.0.2? Tell 10.0.0.1
17	11.365967	Private_66:68:00	Broadcast	ARP	64 Who has 10.0.0.1? Tell 10.0.0.3
18	11.373133	0c:57:62:65:00:00	Private_66:68:00	ARP	60 10.0.0.1 is at 0c:57:62:65:00:00
19	11.374679	10.0.0.3	10.0.0.1	ICMP	98 Echo (ping) request id=0x8532, seq=1/256, ttl=64 (reply in 20)
20	11.377764	10.0.0.1	10.0.0.3	ICMP	98 Echo (ping) reply id=0x8532, seq=1/256, ttl=64 (request in 19)
21	12.378815	10.0.0.3	10.0.0.1	ICMP	98 Echo (ping) request id=0x8632, seq=2/512, ttl=64 (reply in 22)
22	12.379903	10.0.0.1	10.0.0.3	ICMP	98 Echo (ping) reply id=0x8632, seq=2/512, ttl=64 (request in 21)
23	16.487546	0c:57:62:65:00:00	Private_66:68:00	ARP	60 Who has 10.0.0.3? Tell 10.0.0.1
24	16.488342	Private_66:68:00	0c:57:62:65:00:00	ARP	60 10.0.0.3 is at 00:50:79:66:68:00

Рис. 12: Анализ DHCP пакетов в Wireshark

Расширенная топология для настройки DHCPv6

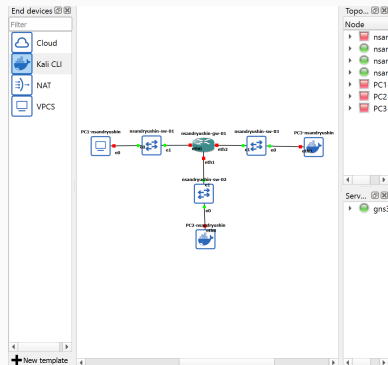


Рис. 13: Расширенная топология для настройки DHCPv6

Настройка IPv6 адресов на интерфейсах

```
nsandryushin@nsandryushin-gw-01# set interfaces ethernet eth1 address 2000::1/64
Configuration path: [interfaces ethernet eth1 address 2000::1/64] already exists
[edit]
nsandryushin@nsandryushin-gw-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
nsandryushin@nsandryushin-gw-01# show interfaces
  ethernet eth0 {
    address 10.0.0.1/24
    hw-id 0c:57:62:65:00:00
  }
+ ethernet eth1 {
+   address 2000::1/64
+   hw-id 0c:57:62:65:00:01
+ }
+ ethernet eth2 {
+   address 2001::1/64
+   hw-id 0c:57:62:65:00:02
+ }
  loopback lo {
  }
[edit]
nsandryushin@nsandryushin-gw-01# commit
[edit]
nsandryushin@nsandryushin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
nsandryushin@nsandryushin-gw-01#
```

Рис. 14: Настройка IPv6 адресов на интерфейсах

Настройка Stateless DHCPv6 на интерфейсе eth1

```
sandryushin@sandryushin-ge-01# set interfaces ethernet eth1 address 2001::1/64
Configuration path: [interface ethernet eth1 address 2001::1/64] already exists
[edit]
sandryushin@sandryushin-ge-01# set interfaces ethernet eth2 address 2001::1/64
[edit]
sandryushin@sandryushin-ge-01# show interfaces
interface eth0 {
    address 10.0.0.1/24
    hw-id 0c157c216510b100
}
interface eth1 {
    address 2001::1/64
    hw-id 0c157c216510b100
}
interface eth2 {
    address 2001::1/64
    hw-id 0c157c216510b100
}
loophack lo {
}
[edit]
sandryushin@sandryushin-ge-01# commit
[edit]
sandryushin@sandryushin-ge-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
sandryushin@sandryushin-ge-01# set service router-advert interface eth1 prefix
2001::/64
[edit]
sandryushin@sandryushin-ge-01# set service router-advert interface eth1 other-
config-file
[edit]
sandryushin@sandryushin-ge-01# set service dhcpv6-server shared-network-name n
sandryushin-stateless
[edit]
sandryushin@sandryushin-ge-01# set service dhcpv6-server shared-network-name n
sandryushin-stateless subnet 2001::/64
[edit]
sandryushin@sandryushin-ge-01# set service dhcpv6-server shared-network-name n
sandryushin-stateless common-options name-server 2001::1
[edit]
sandryushin@sandryushin-ge-01# set service dhcpv6-server shared-network-name n
sandryushin-stateless common-options domain-search sandryushin.net
[edit]
sandryushin@sandryushin-ge-01# commit
[edit]
sandryushin@sandryushin-ge-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
sandryushin@sandryushin-ge-01#
```

Рис. 15: Настройка Stateless DHCPv6 на интерфейсе eth1

Просмотр конфигурации маршрутизатора

```
nsandryushin@nsandryushin-02:~$ cat show configuration
```

```
interface {  
  ethernet eth0 {  
    address 18.0.0.1/24  
    hw-id 0a:83:62:65:00:00  
  }  
  ethernet eth1 {  
    address 2000:1::/64  
    hw-id 0a:83:62:65:00:00  
  }  
  ethernet eth2 {  
    address 2001::1/64  
    hw-id 0a:83:62:65:00:02  
  }  
  loopback lo {  
  }  
}  
  
service {  
  dhcp-server {  
    shared-network-name nsandryushin {  
      domain-name nsandryushin.net  
      name-server 10.0.0.1  
      subnet 10.0.0.0/24 {  
        default-router 10.0.0.1  
      }  
    }  
  }  
}  
...  
interface {  
  ethernet eth0 {  
    address 18.0.0.1/24  
    hw-id 0a:83:62:65:00:00  
  }  
  ethernet eth1 {  
    address 2000:1::/64  
    hw-id 0a:83:62:65:00:00  
  }  
  ethernet eth2 {  
    address 2001::1/64  
    hw-id 0a:83:62:65:00:02  
  }  
  loopback lo {  
  }  
}  
  
service {  
  dhcp-server {  
    shared-network-name nsandryushin {  
      domain-name nsandryushin.net  
      name-server 10.0.0.1  
      subnet 10.0.0.0/24 {  
        default-router 10.0.0.1  
      }  
    }  
  }  
}  
...  
interface {  
  ethernet eth0 {  
    address 18.0.0.1/24  
    hw-id 0a:83:62:65:00:00  
  }  
  ethernet eth1 {  
    address 2000:1::/64  
    hw-id 0a:83:62:65:00:00  
  }  
  ethernet eth2 {  
    address 2001::1/64  
    hw-id 0a:83:62:65:00:02  
  }  
}
```

Рис. 16: Просмотр конфигурации маршрутизатора

Проверка автоконфигурации IPv6 (SLAAC) на PC2

```
-- (root@PC1-neandropushkin: ~)
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet fe80::42:1fff:fea7:6001 prefixlen 64 scopeid 0x25<link>
    inet6 2001::42:1fff:fea7:6001 prefixlen 64 scopeid 0x25<global>
    ether 02:42:11:a7:06:00 txqueuelen 1000 (Ethernet)
    RX packets 13 bytes 1368 (1.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 15 bytes 1162 (1.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet fe80::42:1fff:fea7:6001 prefixlen 64 scopeid 0x25<link>
    ether 02:42:11:a7:06:00 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

-- (root@PC2-neandropushkin: ~)
# route -n -A inet6
Kernel IPv6 routing table

Destination      Next Hop          Flag Met Ref Use If
-----
2001::64          ::                U         256 1   0 eth0
fe80::64          ::                U         256 1   0 eth0
fe80::64          ::                U         256 1   0 eth0
::              fe80::57:62ff:fe05:1  SSMAX 1024 1   0 eth0
::              ::
::1/128          ::                UH        8   3   0 lo
2001::42:2elf:fea7:6000/128 ::                Un        8   2   0 eth0
fe80::42:2elf:fea7:6000/128 ::                Un        8   2   0 eth0
fe80::42:2elf:fea7:601/128  ::                Un        8   3   0 eth0
::              ::
::1/64           ::                U         256 4   0 eth0
::1/64           ::                U         256 1   0 eth0
::1/64           ::                Un        -1  1   0 lo
::1/64           ::

-- (root@PC2-neandropushkin: ~)
# ping 2001::1 -c 2
PING 2001::1 (2001::1) 56 data bytes
64 bytes from 2001::1: icmp_seq=1 ttl=64 time=14.0 ms
64 bytes from 2001::1: icmp_seq=2 ttl=64 time=4.10 ms

-- 2001::1 ping statistics --
2 packets transmitted, 2 received, 0% packet loss, time 100ms
rtt min/avg/max/mdev = 4.892/9.532/14.773/4.940 ms

-- (root@PC2-neandropushkin: ~)
#
```

Рис. 17: Проверка автоконфигурации IPv6 (SLAAC) на PC2

Получение DNS-настроек через Stateless DHCPv6 на PC2

```
(root@ PC2-nsandryushin)-[~]
# cat /etc/resolv.conf

(root@ PC2-nsandryushin)-[~]
# dhclient -6 -s -v eth0
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on Socket/eth0
Sending on Socket/eth0
Created duid "\000\003\000\001\002B\341\247\006\000".
PRC: Requesting information (INIT).
XMT: Forming Info-Request, 0 ms elapsed.
RMT: Info-Request on eth0, interval 930ms.
RCV: Reply message on eth0 from fe80::e57:62ff:fe65:1.
PRC: Done.

(root@ PC2-nsandryushin)-[~]
# ping 2000::1 -c 2
PING 2000::1 (2000::1) 56 data bytes
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=5.59 ms
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=3.30 ms

--- 2000::1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 3.303/4.447/5.592/1.144 ms

(root@ PC2-nsandryushin)-[~]
# cat /etc/resolv.conf
search nsandryushin.net.
nameserver 2000::1

(root@ PC2-nsandryushin)-[~]
```

Рис. 18: Получение DNS-настроек через Stateless DHCPv6 на PC2

Проверка таблицы аренды DHCPv6 (Stateless)

```
nsandryushin@nsandryushin-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
[edit]
nsandryushin@nsandryushin-gw-01#
```

Рис. 19: Проверка таблицы аренды DHCPv6 (Stateless)

Анализ пакетов Stateless DHCPv6 в Wireshark

14	320.484031	2000::42:e1ff:fea7:...	2000::1	ICMPv6	118	Echo (ping) request id=0x0001, seq=1, hop limit=64 (reply in 15)
15	320.487452	2000::1	2000::42:e1ff:fea7:...	ICMPv6	118	Echo (ping) reply id=0x0001, seq=1, hop limit=64 (request in 14)
16	321.475769	2000::42:e1ff:fea7:...	2000::1	ICMPv6	118	Echo (ping) request id=0x0001, seq=2, hop limit=64 (reply in 17)
17	321.479144	2000::1	2000::42:e1ff:fea7:...	ICMPv6	118	Echo (ping) reply id=0x0001, seq=2, hop limit=64 (request in 16)
18	325.816581	fe80::e57:62ff:fe65:...	2000::42:e1ff:fea7:...	ICMPv6	86	Neighbor Solicitation for 2000::42:e1ff:fea7:600 from 0c:57:62:65:00:01
19	325.816977	2000::42:e1ff:fea7:...	fe80::e57:62ff:fe65:...	ICMPv6	78	Neighbor Advertisement 2000::42:e1ff:fea7:600 (sol)
20	330.986795	fe80::42:e1ff:fea7:...	fe80::e57:62ff:fe65:...	ICMPv6	86	Neighbor Solicitation for fe80::e57:62ff:fe65:1 from 02:42:e1:a7:06:00
21	330.987901	fe80::e57:62ff:fe65:...	fe80::42:e1ff:fea7:...	ICMPv6	78	Neighbor Advertisement fe80::e57:62ff:fe65:1 (rtr, sol)
22	336.057030	fe80::e57:62ff:fe65:...	fe80::42:e1ff:fea7:...	ICMPv6	86	Neighbor Solicitation for fe80::42:e1ff:fea7:600 from 0c:57:62:65:00:01
23	336.058004	fe80::42:e1ff:fea7:...	fe80::e57:62ff:fe65:...	ICMPv6	78	Neighbor Advertisement fe80::42:e1ff:fea7:600 (sol)
24	371.973441	fe80::42:e1ff:fea7:...	ff02::1:2	DHCPv6	98	Information-request XID: 0x7b23c6 CID: 000300010242e1a70600
25	371.996563	fe80::e57:62ff:fe65:...	fe80::42:e1ff:fea7:...	DHCPv6	140	Reply XID: 0x7b23c6 CID: 000300010242e1a70600
26	377.016724	fe80::e57:62ff:fe65:...	fe80::42:e1ff:fea7:...	ICMPv6	86	Neighbor Solicitation for fe80::42:e1ff:fea7:600 from 0c:57:62:65:00:01
27	377.016927	fe80::42:e1ff:fea7:...	fe80::e57:62ff:fe65:...	ICMPv6	78	Neighbor Advertisement fe80::42:e1ff:fea7:600 (sol)
28	382.185933	fe80::42:e1ff:fea7:...	fe80::e57:62ff:fe65:...	ICMPv6	86	Neighbor Solicitation for fe80::e57:62ff:fe65:1 from 02:42:e1:a7:06:00
29	382.190584	fe80::e57:62ff:fe65:...	fe80::42:e1ff:fea7:...	ICMPv6	78	Neighbor Advertisement fe80::e57:62ff:fe65:1 (rtr, sol)
30	414.376797	2000::42:e1ff:fea7:...	2000::1	ICMPv6	118	Echo (ping) request id=0x0002, seq=1, hop limit=64 (reply in 31)
31	414.381485	2000::1	2000::42:e1ff:fea7:...	ICMPv6	118	Echo (ping) reply id=0x0002, seq=1, hop limit=64 (request in 30)
32	415.379854	2000::42:e1ff:fea7:...	2000::1	ICMPv6	118	Echo (ping) request id=0x0002, seq=2, hop limit=64 (reply in 33)
33	415.382426	2000::1	2000::42:e1ff:fea7:...	ICMPv6	118	Echo (ping) reply id=0x0002, seq=2, hop limit=64 (request in 32)

Рис. 20: Анализ пакетов Stateless DHCPv6 в Wireshark

Настройка Stateful DHCPv6 на интерфейсе eth2

```
nsandryushin@nsandryushin-gw-01# set service router-advert interface eth2 manage
d-flag
[edit]
nsandryushin@nsandryushin-gw-01# set service dhcpv6-server shared-network-name n
sandryushin-stateful
[edit]
nsandryushin@nsandryushin-gw-01# set service dhcpv6-server shared-network-name n
sandryushin-stateful subnet 2001::0/64
[edit]
nsandryushin@nsandryushin-gw-01# set service dhcpv6-server shared-network-name n
sandryushin-stateful subnet 2001::0/64 name-server 2001::1
[edit]
nsandryushin@nsandryushin-gw-01# set service dhcpv6-server shared-network-name n
sandryushin-stateful subnet 2001::0/64 domain-search nsandryushin.net
[edit]
nsandryushin@nsandryushin-gw-01# set service dhcpv6-server shared-network-name n
sandryushin-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::
199
[edit]
nsandryushin@nsandryushin-gw-01# commit
[edit]
nsandryushin@nsandryushin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

Рис. 21: Настройка Stateful DHCPv6 на интерфейсе eth2

Проверка таблицы аренды перед подключением клиента

```
nsandryushin@nsandryushin-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication    Lease expiration      Remaining
Type      Pool      IAID_DUID
-----
[edit]
nsandryushin@nsandryushin-gw-01#
```

Рис. 22: Проверка таблицы аренды перед подключением клиента

Начальное состояние сети на PC3

```
(root@PC3-nsandryushin) ~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet fe80::42:dcff:fe52:8600 prefixlen 64 scopeid 0x20<link>
    ether 02:42:dc:52:86:00 txqueuelen 1000 (Ethernet)
    RX packets 17 bytes 1070 (1.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 15 bytes 1146 (1.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet fe80::42:dcff:fe52:8601 prefixlen 64 scopeid 0x20<link>
    ether 02:42:dc:52:86:01 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(root@PC3-nsandryushin) ~# route -n -A inet6
Kernel IPv6 routing table

```

Destination	Next Hop	Flag	Met	Ref	Use	If
fe80::/64	::	U	256	1	0	eth0
fe80::/64	::	U	256	1	0	eth1
::/0	fe80::e57:62ff:fe65:2	UG	1024	1	0	eth0
::/128	::	Un	0	2	0	lo
fe80::42:dcff:fe52:8600/128	::	Un	0	3	0	eth0
fe80::42:dcff:fe52:8601/128	::	Un	0	2	0	eth1
ff00::/8	::	U	256	4	0	eth0
ff00::/8	::	U	256	1	0	eth1
::/0	::	Un	-1	1	0	lo

```
(root@PC3-nsandryushin) ~# cat /etc/resolv.conf
```

Рис. 23: Начальное состояние сети на PC3

Получение адреса через Stateful DHCPv6 на PC3

```
---(root@PC3-aaa000000000)---
root@PC3:~# dhclient -6 -v eth0
Internet Systems Consortium DHCP Client 4.4.1-P1
Copyright 2004-2012 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on Socket/eth0
Sending on Socket/eth0
Created uid "00000010000001000000\377A\002B\334\206\000".
MSG: Soliciting for leases (INIT).
MSG: Forming Solicit, 0 ms elapsed.
MSG: X-- IA NA d:152:86:00
MSG: I X-- Request renew in +3600
MSG: I X-- Request rebind in +5400
MSG: Solicit on eth0, interval 100ms.
RCV: Advertise message on eth0 from fe80::e57:62ff:fe6b::2.
RCV: X-- IA NA d:152:86:00
RCV: I X-- starts 1765032642
RCV: I X-- t1 - renew +0
RCV: I X-- t2 - rebind +0
RCV: I X-- [Options]
RCV: I X-- IAADDR 2001::139
RCV: I I X-- Preferred lifetime 27000.
RCV: I I X-- Max lifetime 43200.
RCV: X-- Server ID: 00:01:00:01:30:c6:fa:39:0c:57:62:65:00:01
RCV: Advertisement recorded.
MSG: Selecting best advertised lease.
MSG: Considering best lease.
MSG: X-- Initial candidate 00:01:00:01:30:c6:fa:39:0c:57:62:65:00:01 (s: 10105, p: 0).
MSG: Forming Request, 0 ms elapsed.
MSG: X-- IA NA d:152:86:00
MSG: I X-- Requested renew +3600
MSG: I X-- Requested rebind +5400
MSG: I X-- IAADDR 2001::139
MSG: I I X-- Preferred lifetime +7200
MSG: I I X-- Max lifetime +7500
MSG: V IA NA appended.
MSG: Request on eth0, interval 100ms.
RCV: Reply message on eth0 from fe80::e57:62ff:fe6b::2.
RCV: X-- IA NA d:152:86:00
RCV: I X-- starts 1765032643
RCV: I X-- t1 - renew +0
RCV: I X-- t2 - rebind +0
RCV: I X-- [Options]
RCV: I X-- IAADDR 2001::139
RCV: I I X-- Preferred lifetime 7200.
RCV: I I X-- Max lifetime 7500.
RCV: X-- Server ID: 00:01:00:01:30:c6:fa:39:0c:57:62:65:00:01
MSG: Bound to lease 00:01:00:01:30:c6:fa:39:0c:57:62:65:00:01.
```

Рис. 24: Получение адреса через Stateful DHCPv6 на PC3

Проверка сетевых настроек на РС3 после работы DHCPv6

[illegible]

Рис. 25: Проверка сетевых настроек на PC3 после работы DHCPv6

Проверка таблицы аренды DHCPv6 (Stateful) с активной записью

```
nsandryushin@nsandryushin-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type              Pool                               IAID_DUID
-----
-----
2001::199         active    2025/12/06 14:50:43      2025/12/06 16:55:43    2:02:28
non-temporary    nsandryushin-stateful  00:86:52:dc:00:01:00:01:30:c6:ff:41:02:42
:dc:52:86:00
[edit]
nsandryushin@nsandryushin-gw-01#
```

Рис. 26: Проверка таблицы аренды DHCPv6 (Stateful) с активной записью

Анализ пакетов Stateful DHCPv6 в Wireshark

1 0.000000	fe80::e57:62ff:fe65::	ff02::16	ICMPv6	150 Multicast Listener Report Message v2
2 0.003582	fe80::e57:62ff:fe65::	ff02::16	ICMPv6	170 Multicast Listener Report Message v2
3 0.332013	::	ff02::1:ff00:1	ICMPv6	86 Neighbor Solicitation for 2001::1
4 0.595706	fe80::e57:62ff:fe65::	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
5 484.617556	fe80::42:dcff:fe52::	ff02::2	ICMPv6	70 Router Solicitation from 02:42:dc:52:86:00
6 937.575044	fe80::e57:62ff:fe65::	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
7 937.585840	fe80::e57:62ff:fe65::	ff02::16	ICMPv6	110 Multicast Listener Report Message v2
8 938.337732	fe80::e57:62ff:fe65::	ff02::16	ICMPv6	90 Multicast Listener Report Message v2
9 940.186034	fe80::e57:62ff:fe65::	ff02::1	ICMPv6	86 Router Advertisement from 0c:57:62:65:00:02
10 957.068438	fe80::e57:62ff:fe65::	ff02::1	ICMPv6	86 Router Advertisement from 0c:57:62:65:00:02
11 973.089872	fe80::e57:62ff:fe65::	ff02::1	ICMPv6	86 Router Advertisement from 0c:57:62:65:00:02
12 1488.232834	fe80::42:dcff:fe52::	ff02::1:2	DHCPv6	118 Solicit XID: 0x779f36 CID: 0001000130c6ff410242dc528600
13 1488.308014	fe80::e57:62ff:fe65::	fe80::42:dcff:fe52::	DHCPv6	188 Advertise XID: 0x779f36 IAA: 2001::199 CID: 0001000130c6ff410242dc528600
14 1489.284769	fe80::42:dcff:fe52::	ff02::1:2	DHCPv6	164 Request XID: 0x19720c CID: 0001000130c6ff410242dc528600 IAA: 2001::199
15 1489.349186	fe80::e57:62ff:fe65::	fe80::42:dcff:fe52::	DHCPv6	188 Reply XID: 0x19720c IAA: 2001::199 CID: 0001000130c6ff410242dc528600
16 1489.451784	fe80::42:dcff:fe52::	ff02::16	ICMPv6	110 Multicast Listener Report Message v2
17 1489.555528	::	ff02::1:ff00:199	ICMPv6	86 Neighbor Solicitation for 2001::199
18 1489.785466	fe80::42:dcff:fe52::	ff02::16	ICMPv6	110 Multicast Listener Report Message v2
19 1493.739756	fe80::e57:62ff:fe65::	fe80::42:dcff:fe52::	ICMPv6	86 Neighbor Solicitation for fe80::42:dcff:fe52:8600 from 0c:57:62:65:00:02
20 1493.740225	fe80::42:dcff:fe52::	fe80::e57:62ff:fe65::	ICMPv6	78 Neighbor Advertisement fe80::42:dcff:fe52:8600 (sol)
21 1498.889314	fe80::42:dcff:fe52::	fe80::e57:62ff:fe65::	ICMPv6	86 Neighbor Solicitation for fe80::e57:62ff:fe65:2 from 02:42:dc:52:86:00
22 1498.890765	fe80::e57:62ff:fe65::	fe80::42:dcff:fe52::	ICMPv6	78 Neighbor Advertisement fe80::e57:62ff:fe65:2 (rtr, sol)
23 1538.201445	fe80::e57:62ff:fe65::	ff02::1	ICMPv6	86 Router Advertisement from 0c:57:62:65:00:02
24 1606.244819	2001::199	2001::1	ICMPv6	118 Echo (ping) request id=0x0003, seq=1, hop limit=64 (reply in 27)
25 1606.249346	2001::1	ff02::1:ff00:199	ICMPv6	86 Neighbor Solicitation for 2001::199 from 0c:57:62:65:00:02
26 1606.249660	2001::199	2001::1	ICMPv6	86 Neighbor Advertisement 2001::199 (sol, ovr) is at 02:42:dc:52:86:00
27 1606.251028	2001::1	2001::199	ICMPv6	118 Echo (ping) reply id=0x0003, seq=1, hop limit=64 (request in 24)
28 1607.248246	2001::199	2001::1	ICMPv6	118 Echo (ping) request id=0x0003, seq=2, hop limit=64 (reply in 29)
29 1607.252853	2001::1	2001::199	ICMPv6	118 Echo (ping) reply id=0x0003, seq=2, hop limit=64 (request in 28)
30 1611.530679	fe80::42:dcff:fe52::	2001::1	ICMPv6	86 Neighbor Solicitation for 2001::1 from 02:42:dc:52:86:00
31 1611.530889	fe80::42:dcff:fe52::	fe80::e57:62ff:fe65::	ICMPv6	86 Neighbor Solicitation for fe80::e57:62ff:fe65:2 from 02:42:dc:52:86:00
32 1611.536343	2001::1	fe80::42:dcff:fe52::	ICMPv6	78 Neighbor Advertisement 2001::1 (rtr, sol)
33 1611.536426	fe80::e57:62ff:fe65::	fe80::42:dcff:fe52::	ICMPv6	78 Neighbor Advertisement fe80::e57:62ff:fe65:2 (rtr, sol)

Рис. 27: Анализ пакетов Stateful DHCPv6 в Wireshark

В результате выполнения лабораторной работы были получены навыки использования и настройки dhcpv6